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Patient Path

“Your medical story, wherever you go”

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Who We Are

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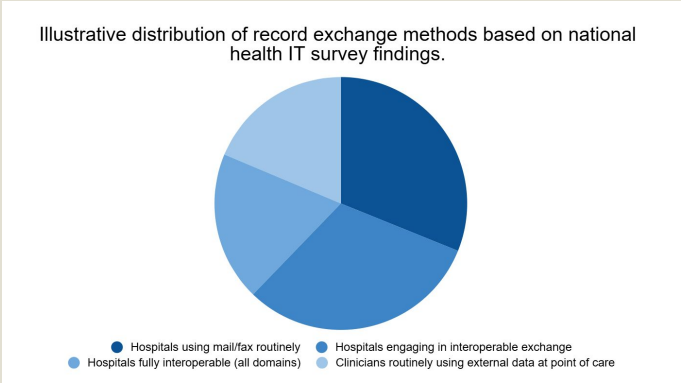
Elevator Pitch

PatientPath is a secure web-based platform that replaces outdated fax systems in healthcare with instant digital record sharing. It allows providers to access and share patient records in real time while integrating with existing EHR (electronic health record) systems. By reducing delays, errors, and administrative work, PatientPath helps deliver faster, safer, and more coordinated care, so patients' medical stories follow them wherever they go.

The Problem

Current medical transfer methods are inefficient and slow.

- 65% of all transfers rely on fax machines despite the availability of digital alternatives
- Slow, unreliable, and prone to errors
  - Issues include lost pages, unreadable documents, and delayed delivery
- Direct impact on patient care
  - Leads to postponed diagnosis and repeated tests
- Disproportionate Harm
  - Especially impactful in rural and underfunded facilities



Office of the National Coordinator for Health Information Technology. (2023). *Interoperable exchange of patient health information among U.S. hospitals: 2023 data brief.* U.S. Department of Health & Human Services. https://www.healthit.gov/data/data-briefs/interoperable-exchange-patient-health-information-among-us-hospitals-2023

Current State of Healthcare IT

Although some healthcare practices have implemented EHR systems, **interoperability still remains a major challenge**.
Key issues include:

- Many EHR systems do not communicate across different organizations.
- Healthcare IT systems remain siloed.
- Smaller and rural communities often lack modern infrastructure.
- Regulatory requirements (HIPAA) increase complexity.

As a result, when systems cannot exchange data directly, providers often default to manual or outdated communication methods.

Source: Office of the National Coordinator for Health Information Technology (2023). Interoperable exchange of patient health information among U.S. hospitals.

Why This Matters

Healthcare is very time sensitive. Delays in information transfer can lead to:

- Postponed specialist appointments
- Repeated diagnostic testing
- Incomplete patient histories
- Increased workload for administrative staff
- All around frustration for patients and providers

In an industry where minutes can affect outcomes, inefficiencies in information exchange have real clinical and financial consequences.

Problem Characteristics

Patient Privacy Risk

Outdated security protocols

Confidential data vulnerability

Disconnected Systems

Limited interoperability

Inefficient Transfer Process

Record delays

Document loss

Illegible fax transmissions

Problem Characteristics

Scalability Barriers

Limited growth adaptability

Rigid architecture

Poor Usability

Cluttered interfaces

Poor navigation

Lack of user roles

Connectivity Constraints

Online-only architecture

Downtime vulnerability

Who is Affected?

- Patients
- Providers
- Admin Staff
- Rural Facilities
- Families/Caregivers

Current Flow Process

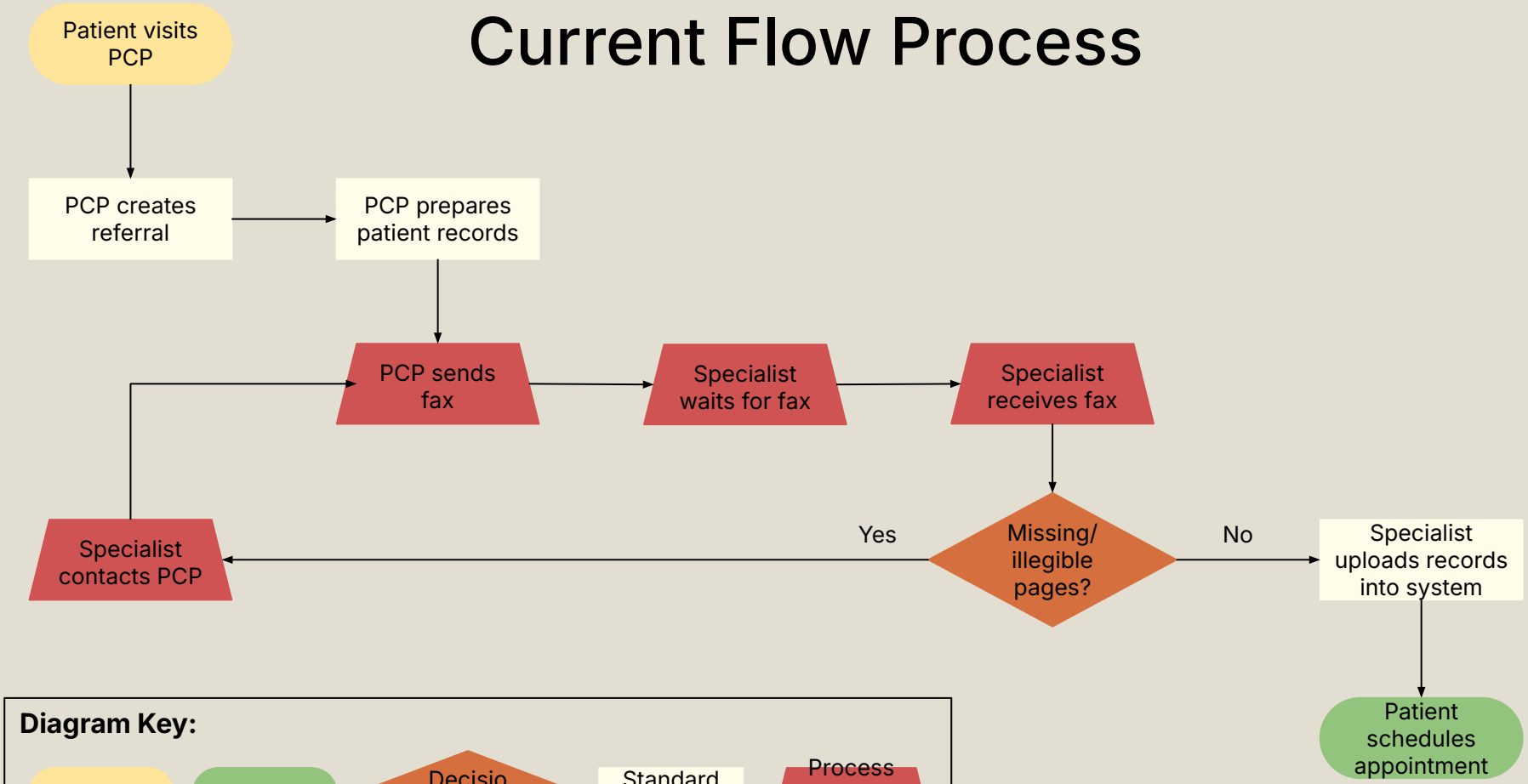


Diagram Key:

Start | Stop | Decision | Standard Step | Process Bottleneck

The Solution

PatientPath is a secure, centralized digital platform designed to replace outdated fax-based communication in healthcare.

Instead of faxing documents, PatientPath enables providers to instantly upload, access, and share medical records through a secure web-based system.

The platform supports hospitals, clinics, private practices, healthcare networks, and their patients.

Unlike traditional EHR systems, PatientPath focuses on secure interoperability and cross-organization communication. It acts as a **secure bridge between systems**, reducing reliance on fax while improving cybersecurity, compliance, and usability.

Solution Characteristics

Security & Compliance

- End-to-end encryption (data in transit + at rest)
- HIPAA-aligned architecture
- Role-Based Access Control (organization-level permissions)
- Multi-factor authentication
- Full audit trail logging

Solution Characteristics

Interoperability & Architecture

- API-first REST architecture
- FHIR & HL7 compatibility
- EHR-agnostic integration (Epic, Oracle, etc.)
- Plugin-based deployment model

Solution Characteristics

Performance & Scalability

- Cloud-hosted (AWS/Azure/GCP compatible)
- 99.9%+ uptime target
- Horizontal scaling architecture
- Offline sync resilience

Major Functional Components

Identity and Access Management

- Secure authentication with multi-factor Authentication (MFA)
- Role Based Access Control (RBAC) with organization-level permissions
- Provider and patient identity verification

Patient Record Management

- Secure Document upload, tagging, and version control
- Indexing, search, and duplicate detection
- Full audit logging and consent tracking

Interoperability and EHR Integration

- API- first architecture with FHIR/HL7 compliance
- EHR system connectors and data normalization
- Secure data routing and plugin support

Major Functional Components

Secure Record Exchange

- Direct provider-to-provider and cross-organization transfer
- Transfer request workflow with approval process
- Real-time tracking and delivery confirmation

Notifications

- Automated alerts for referrals, status, and deadlines
- Event- driven triggers and provider messaging

Provider Dashboard and Patient Portal

- Centralized interfaced for referrals, requests, and monitoring
- Patient record access and consent management
- Organization contacts and transfer history

Problem–Solution Mapping

Identified Problem	System Feature	Expected Impact
Reliance on fax-based record transfers	Secure real-time digital document exchange	Reduced transfer delays and elimination of document loss
Limited interoperability between EHR systems	API-first architecture with FHIR/HL7 compatibility	Standardized cross-organization data exchange
Manual referral tracking and follow-up processes	Automated referral workflow with status monitoring	Improved operational efficiency and faster referral completion
Communication methods with security vulnerabilities	End-to-end encryption with RBAC and MFA	Enhanced HIPAA-aligned compliance and access control
Connectivity instability in rural or under-resourced facilities	Offline caching with automatic synchronization	Sustained system usability in low-bandwidth environments

Patient Path Will

What Patient Path Will Do

- Enable secure digital transfer of medical records between providers
- Provide a centralized hub for accessing and managing patient documents
- Offer real-time tracking of record requests and transfer status
- Support digital patient consent and authorization workflows
- Reduce errors with automated validation and completeness checks
- Maintain HIPAA-compliant audit trails and access controls

Patient Path Will Not

Provide medical diagnoses or clinical recommendations - The platform does not assist in medical decision making. It’s role is limited to securely transferring and accessing information.

Handle hospital management functions - Scheduling, billing, and internal hospital operations are already handled by specialized systems.

Function as an emergency communication system - The platform supports data exchange, not urgent/emergency situations.

Solution Process Flow

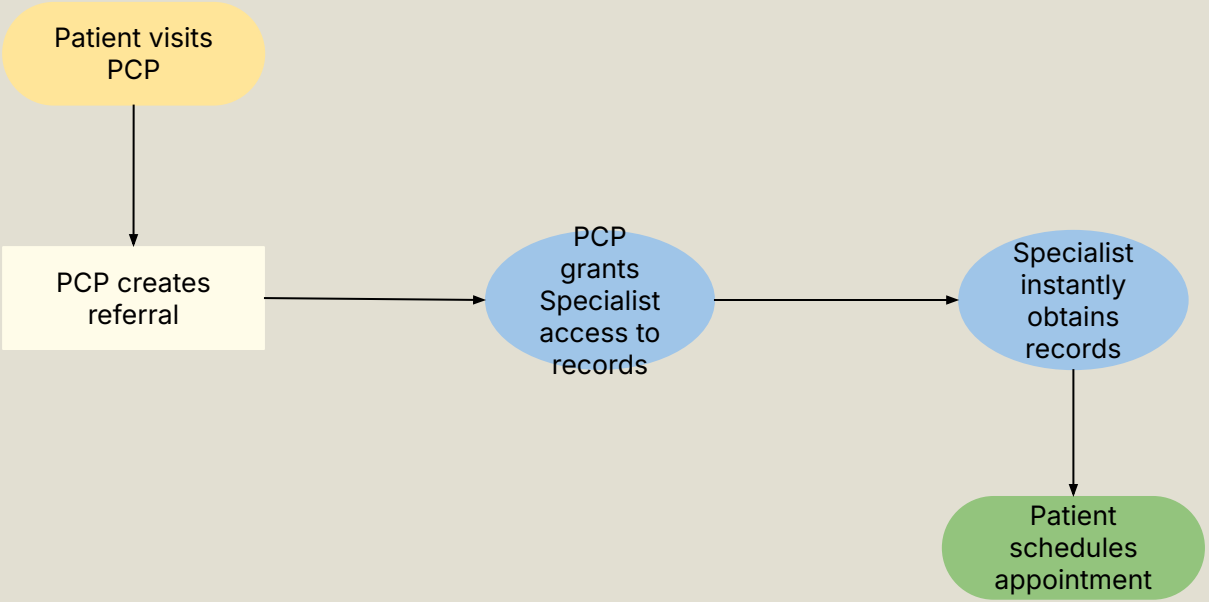


Diagram Key:

Risks

Risk	Likelihood	Impact	Mitigation
EHR integration complexity	High	High	FHIR APIs, phased rollout
Regulatory compliance (HIPAA)	Medium	High	Audits, legal oversight
Provider adoption resistance	Medium	High	Training, workflow alignment
Cybersecurity breach	Medium	High	Encryption, MFA
Data transfer errors	Medium	High	Validation checks
Patient trust concerns	Medium	Medium	Transparent consent

Market Assessment

Target Segments

- Hospitals and Healthcare networks
- Independent private practices
- Specialty Clinics

Competitive Landscape

- Traditional fax transfers
- Electronic Health Record systems (Epic, Oracle Healthcare, and eClinicWorks)

Conclusion

References

Office of the National Coordinator for Health Information Technology. (2023). *Interoperable exchange of patient health information among U.S. hospitals: 2023 data brief*. U.S. Department of Health & Human Services. https://www.healthit.gov/data/data-briefs/interoperable-exchange-patient-health-information-among-us-hospitals-2023

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Glossary

PCP - Primary Care Provider
EHR - Electronic Health Records
HIPAA - Health Insurance Portability and Accountability Act
FHIR - Fast Healthcare Interoperability Resources